MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

1/ TYPE DOD-029

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-T.

DESCRIPTION: Coaxial, pulse type, integral magnet, tunable frequency 9,050 to 10,000 MHz, 130 kw minimum peak power output

ABSOLUTE RATINGS:

Parameter:	Ef	if(surge)	tk	epy	ib	Pi	pi	tpc	rrv	Du
Unit:	V	a	sec	kv	a	W	kw	µs	kv/µs	---
Maximum:	13.7	10	---	18.7	20	350	350	4.0	190	0.0011
Minimum:	---	---	180	16.7 (Note 1)	14	---	---	0.1 (Note 1)	50 (Note 2)	---

ABSOLUTE RATINGS:

Parameter:	T(body)	T (input bushing)	Tuner stop torque	Pressurization Input and output	Alt
Unit:	°C	°C	in.-oz	psia	ft
Maximum:	+125	+185	50	45	5,000
Minimum:	-55 (Notes e and f)	-55	20	14 (Note 3)	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Marking:	See note 4
Mounting position:	Any	Cooling:	Forced air (see note 5)
Mounting support:	Flange (see figure 1)	Handling:	See note 6
Coupling:	See figure 1	Weight:	14 pounds (approx)

TEST CONDITIONS: See note 2

Parameter:	Ef	tk	tpc	rrv	Du	Ib	VSWR	F
Unit:	V	sec	µs	kv/µs	---	mAdc	---	MHz
Tolerance:	---	(max)	---	(min)	---	---	(max)	---
Test condition 1:	7.5	180	0.8 ±0.1	160	0.00075	13.5	1.1:1	F1, F2, F3
Test condition 2:	10.5 (Note 8)	180	0.20 ±0.03 (Note 1)	190 (Note 2)	0.00027	4.87	1.1:1	F1, F2, F3

Frequency
F1 = 9,050 MHz
F2 = 9,525 MHz
F3 = 10,000 MHz

GENERAL:

First article sample inspection - Required (see note 9)

1/ Replaces Raytheon Company type number QKH1693 and Cardion Electronics part number 402100

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	First article inspection							
1042	Shock, specified pulse	10	-	Test condition J	---	---	---	---
4223	Tuner stop endurance	-	-	50 in.-oz	---	---	---	---
4223	Mechanical tuning	11,12	-		---	10,000	---	Cycles
4223	Operating torque	-	-	T = -55° and +125°C	Torque	---	20	in.-oz
4027	Temperature coefficient	13,e,f	1	F1 and F3; T(anode) = 70° to 100°C	ΔF/ΔT	---	0.25	MHz/°C
1143	Forced cooling	5,e,f	1	F2	ΔT	---	50	°C
	Quality conformance inspection, part 1	14	-					
4003	Pressurizing	3	-	40 to 45 psia input to output	---	---	---	---
4289	Heater current (nonoperating)	-	-	Ef = 12.6 Vac; t = 180	If	2.0	2.6	A
4223	Operating torque	-	1	T = 25° ±5°C	Torque	---	20	in.-oz
4223	Mechanical tuning range	- -	1 1	Upper limit Lower limit	F F	10,000 ---	--- 9,050	MHz MHz
4223	Resettability	-	1	F2	F	---	2.5	MHz
4223	Tuning rate	15	1		---	145	155	Turns
4306	Pulse voltage	-	1		epy	16.7	18.7	kv
4250	Power output (1)	-	1		Po	97.5	---	W
4250	Power output (2)	-	2		Po	35	---	W
4308	RF bandwidth	16	1,2		BW	---	2.0/tpc	MHz
4308	Minor lobe ratio	16,17	1		SL	8	---	dB
4315	Stability (1)	18	1		MP	---	0.25	%
4315	Stability (2)	18	2		MP	---	0.5	%
4315	Starting stability	18,19,20	2		MP	---	1.0	%
4318	Time jitter	-	2		tj	---	6.0	ns
	Quality conformance inspection, part 2							
1032	Vibration, mechanical	21	1	F3	ΔF MP	--- ---	±2 0.5	MHz %
4266	Direct-interelectrode capacitance	-	-		C	10	14	pF
4310	Frequency pulling figure	22	1		ΔF	---	7	MHz
4311	Frequency pushing figure	23	1	F2	ΔF/ΔA	---	0.225	MHz/A
---	Tuning characteristics	24	1		ΔF	---	±10	MHz

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3							
4551	Intermittent life	7,25	-	Group D; VSWR = 1.5:1 minimum cycled through λg every 30 minutes starting at F1 and increasing in 100 MHz increments every 24 hours for the duration of the life test.	---	1,000	---	Cycles
---	Intermittent life-test end points:							
4250	Power output (1)	-	1		Po	78	---	W
4250	Power output (2)	-	2		Po	28	---	W
4308	RF bandwidth	16	1		BW	---	2.5/tpc	MHz
4315	Stability (1)	18	1		KP	---	0.5	%
---	Tuning characteristics	24	1		ΔF	---	±10	MHz
4223	Resettability	-	1	F2	F	---	4	MHz

NOTES:

1. The characteristics of the applied pulse must be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. The cognizant service electron tube group should be consulted concerning pulse characteristics are related to the specified application.

2. The prv shall be expressed in kV per μs defined by the steepest tangent to the leading edge of the voltage pulse above 70 percent amplitude. Any capacitance used in viewing system shall not exceed 6.0 pF. No spike or ripple shall exceed ±7 percent of the average peak value of voltage or current. Inverse voltage shall not exceed 20 percent of the forward voltage. The negative amplitude of any post voltage oscillation shall not exceed 10 percent of the average peak value.

3. The seal formed by clamping the TUT input and output flanges against suitable magnetron test fixtures shall be hermetically tight for 1-minute minimum with air pressure as specified applied internally to the test fixtures. (See figure 1.)

4. In addition to regular markings, the tuner calibrations, dial readings versus frequency, shall be marked on the tube in the following manner. Note that tuner calibrations should be taken with the tube operating in accordance with test condition 1 conditions and tuning is to be accomplished in one direction only (from low to high frequency).

Frequency (MHz)	Dial reading
9,050	---
9,525	---
10,000	---

5. An airflow of 25 cubic feet per minute (cfm) at approximately 760 mmHg shall be directed on the cooling fins from an orifice 5.2 by 1.0 inches (132.08 by 25.4 mm) located 2 inches (50.8 mm) from the axis of the cathode terminal. Temperature of the anode shall be measured and the rise shall not exceed the amount specified. Back pressure is measured in the air supply duct (at least 2 inches (50.8 mm) back from the tube radiation cooling fins) shall not be greater than 7.6 mm of mercury.

6. In handling and mounting the tube, care shall be exercised to prevent demagnetization. A minimum distance of 9 inches shall be maintained between the magnets and any other ferromagnetic material.

7. If during life test the TUT does not meet the specified limits, it shall be recycled for an additional 2 cycles. At such time, the tests shall be repeated. If the TUT fails the second test, it shall be considered unsatisfactory.

8. Prior to the application of high voltage, the cathode shall be heated to the required initial operating temperature. This may be done by applying 12.6 volts ±5 percent for 3 minutes. On the application of anode power, the heater voltage should be lowered to the voltage specified. The tube heater shall be protected against arcing by the use of a connector that places a minimum capacitance of 4,000 pF across the heater directly at the input terminals.

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NOTES: -Continued

9. First article sample inspection shall conform to the requirements of appendix F of MIL-E-1 and shall consist of performing all tests specified on this tube specification sheet. Invitation for bids should provide that the preparing activity reserves the right to waive the requirements for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

10. Upon completion of the shock tests, the TUT shall meet the power output and stability requirements of quality conformance inspection, part 1.

11. One cycle is defined as a complete traverse from one end of the tuning range to the other and back again. During this test, the tuner may not be lubricated except at start of test. Tuner shaft shall be driven between 325 rpm (min) and 500 rpm (max).

12. After the specified number of cycles, the TUT shall pass the power output, stability, tunable frequency requirements under test condition 1 in quality conformance inspection, part 1. Tuner resettability shall be allowed to degrade to ±4 MHz.

13. This measurement shall be made only after thermal equilibrium has been reached.

14. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

15. With the tuning dial adjusted for a frequency of 9,050 MHz, an operating frequency of 10,000 MHz shall be obtained by counter-clockwise rotation of the tuning shaft (see figure 1) 150 ±5 turns. This test shall be performed in a manner which excludes backlash.

16. The specification limits shall not be exceeded at the point of maximum degradation determined by observation while varying a mismatch of 1.5:1 through 180 degrees.

17. A suitable spectrum is considered one in which the major lobe has a shape such that its slope does not change sign more than once for power levels greater than the specified dB below its peak.

18. Stability shall be measured in terms of the average number of output pulses missing, expressed as a percent of the number of input pulses applied during the period of observation. Pulses are considered to be missing if the rf energy within a 1 percent band about the operating frequency is less than 70 percent of the normal energy level of a pulse. The VSWR shall be adjusted to the phase producing maximum instability and the MP shall be counted for a period of three minutes.

19. The starting stability test shall be performed after a nonoperating holding period of 168 hours minimum.

20. The MP count shall be for a period of 3 minutes starting 30 seconds after application of high voltage.

21. The TUT shall be mounted in a resonance-free jig and vibration with sinusoidal excitation in each of three mutually perpendicular planes through the following amplitude: 5 to 25 Hz at 0.125 DA. Duration of one cycle (5 to 25 to 5 Hz) shall be approximately 3 minutes. Repeat for a total of 30 minutes in each of the three planes. If a resonance is observed, the TUT shall be given an additional two hours vibration at the resonant frequency. If more than one resonance is observed, the test shall be performed at the frequency showing the maximum degradation. Interpulse frequency stability and rf stability shall not exceed the limits specified. Upon completion of the vibration test, the TUT shall meet all of the requirements of quality conformance inspection, part 1, except starting stability.

22. The frequency deviation (maximum frequency minus minimum frequency) shall not exceed the limits specified herein when a VSWR of 1.5:1 minimum, intoduced into the load at a distance of 1 ±0.05 meters from the tube flange, is varied throughout all phases.

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NOTES: -Continued

23. The pushing factor shall be determined in the current range 17 to 19 amperes, and the value shall not exceed the limit specified herein. The peak current through the tube shall alternately be the limits as specified under this test condition. The test shall be run to exclude the effects of thermal drift and frequency instability not due to pushing.

24. Tuning curve linearity is determined in the following manner. A plot of frequency versus dial readings shall be made over the frequency range F1 to F3. Data points shall be obtained for approximately each 100 MHz increment of frequency. To exclude backlash effects, tuning shall be accomplished in one direction only (from low to high frequency). Room ambient conditions 20° to 30°C shall prevail.

25. Life test shall be conducted in accordance with the following cycle:

Test condition	Ib	Ef	Duration
Standby	0	12.6	3.0 minutes
1	13.5 mA	7.5	30 minutes
2	4.87 mA	10.5	30 minutes
Off	0	0	12 minutes

26. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-N085)

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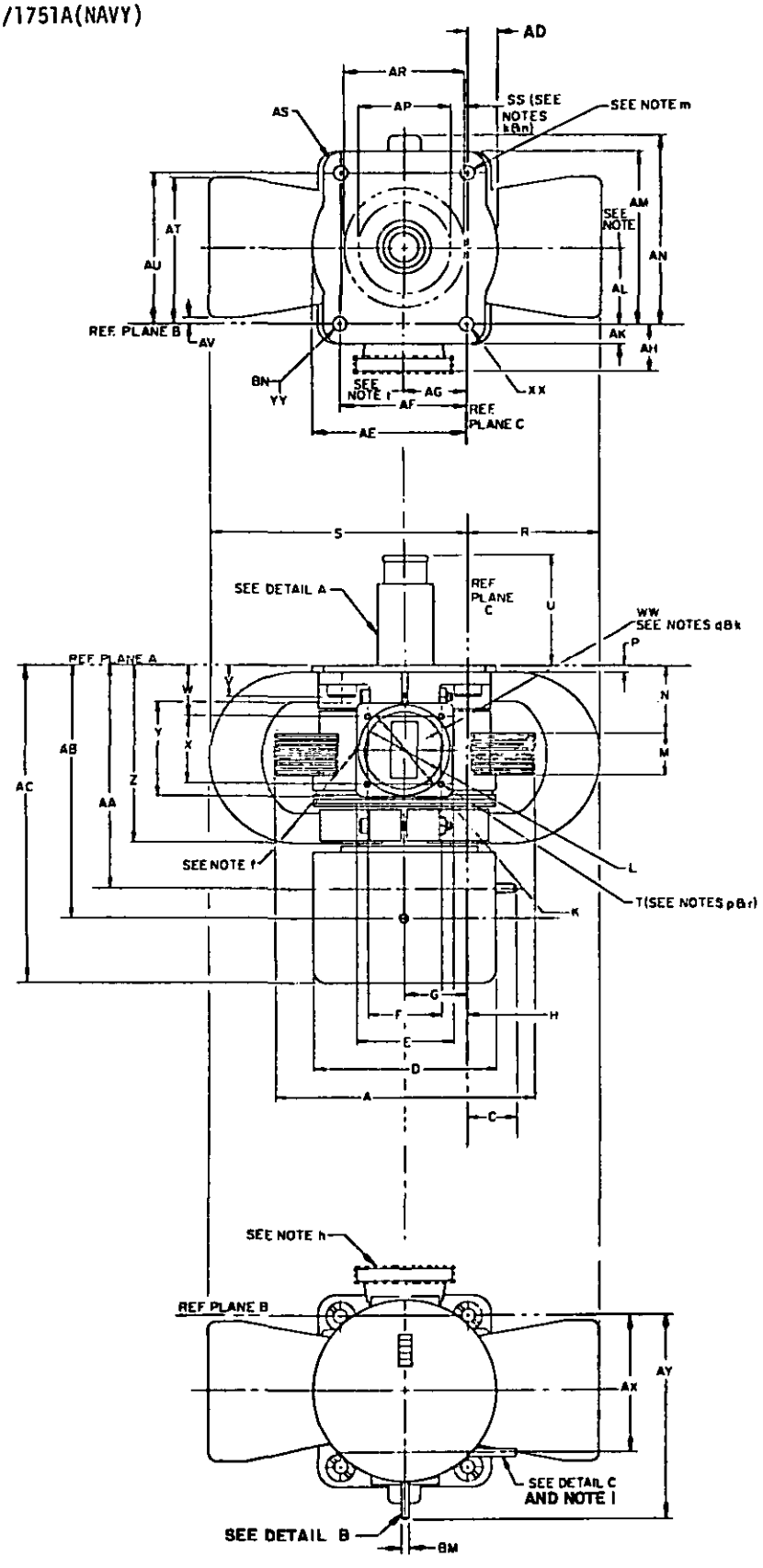
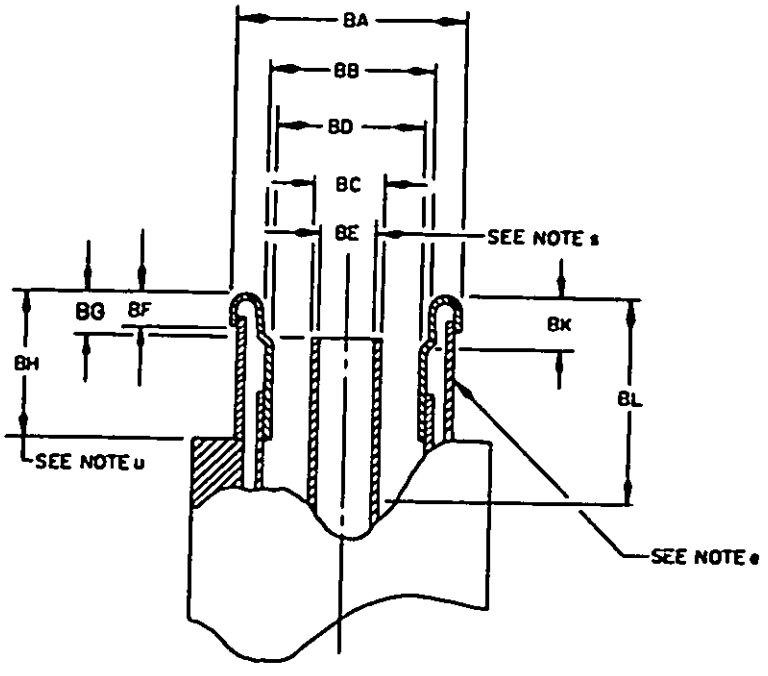
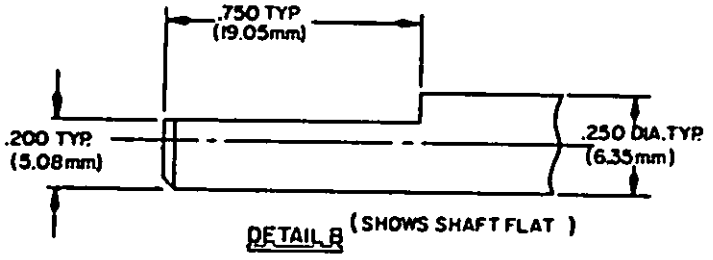


FIGURE 1. Outline drawing of electron tube type DOD-029.

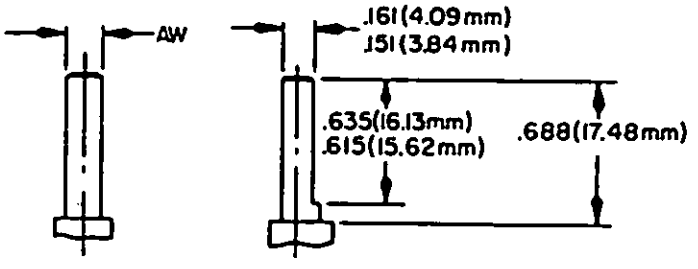
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DETAIL A (SEE NOTE g)



DETAIL B (SHOWS SHAFT FLAT)



DETAIL C (TUNER SHAFT DRIVE)

FIGURE 1. Outline drawing of electron tube type DOD-029-Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	First article inspection	
H	.453 (11.50)	.573 (14.55)
P	.110 (2.79)	.140 (3.56)
T	.164 (4.17) -32 UNC-2B (4 holes)	
V	.545 (13.84)	.645 (16.38)
Y	1.860 (47.24)	1.890 (48.00)
AD	--- ---	.625 (15.88)
AV	.060 (1.52)	--- ---
AW	.1865 (4.74)	.1895 (4.81) Dia.
BB	.605 (15.37)	.619 (15.72)
	Quality conformance inspection, part 1 (see note 14)	
D	---	3.750 (95.25)
E	1.860 (47.24)	1.890 (48.00)
F	1.464 (37.18)	1.484 (37.69)
R	---	2.710 (68.83)
S	---	5.210 (132.33)
U	1.789 (45.44)	1.975 (50.17)
Z	---	3.560 (90.42)
AA	4.378 (111.20)	4.628 (117.55)
AC	---	6.500 (124.25)
AE	---	3.125 (79.38)
AF	2.490 (63.25)	2.510 (63.75)
AM	---	3.459 (87.86)
AN	---	3.900 (99.06)
AT	---	2.937 (74.60)
AU	2.990 (75.95)	3.010 (76.45)
AX	2.627 (66.73)	2.813 (71.45)
BA	.822 (20.88)	.837 (21.26)
BC	.234 (5.94)	.266 (6.76)
BD	.533 (13.54)	.547 (13.89)
BE	.164 (4.17)	.174 (4.42)
BF	.115 (2.92)	.135 (3.43)
BG	.112 (2.84)	.212 (5.38)
BH	.530 (13.46)	---
BL	.765 (19.43)	---
BM	.250 (6.35) Dia.	---
	Quality conformance inspection, part 2	
A	5.058 (128.47)	5.178 (131.52)
C	.817 (20.75)	1.068 (27.13)
M	--- ---	.900 (22.86)
N	1.235 (31.37)	1.421 (36.09)
W	.917 (23.29)	.997 (25.32)
X	1.342 (34.09)	1.362 (34.59)
AB	4.857 (123.37)	5.107 (129.72)
AH	.895 (22.73)	1.015 (25.78)
BK	--- ---	.155 (3.94)
BN	.276 (7.01) Dia. (4 holes)	.286 (7.26)

Nominal dimensions		
Ltr	INCHES	MM
G	1.250	(31.75)
K	1.812	(46.02)
L	1.531	(38.89)
AG	1.250	(31.75)
AK	0.437	(11.10)
AL	1.500	(38.10)
AP	1.800	(45.72)
AR	2.450	(62.23)
AS	0.375R	(9.53)
AY	4.305	(109.36)

FIGURE 1. Outline drawing of electron tube type DOD-029- Continued.

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Notes:

a. Reference plane "A" passes along the face of the mounting plate as shown.
b. Reference plane "B" is perpendicular to plane "A" and passes thru the center of holes "XX" and "YY" at plane "A".
c. Reference plane "C" is mutually perpendicular to planes "A" and "B" and passes thru the center of hole "XX" at plane "A".
d. Mates with modified choke flange, UG-52B/U per MIL-F-3922/59A (clearance instead of threaded holes).
e. Input bushing temperature measured at this point.
f. Anode temperature measured at this point (joint between output and body).
g. Input connector mates with Jettron Products Inc. part no. 90-006 and no. 90-030, or equivalent.
h. Protective cover for shipping purposes only. Remove before operating tube.
k. The waveguide flange seal area "WW" and mounting plate seal area "SS" and assembly joints within these areas must be capable of providing a hermetic seal.
l. Counter-clockwise rotation of tuner shaft at this end increases tube frequency.
m. Axis of hole to be centered on plane "C" within .010 (.254 mm).
n. This area on face of mounting plate to be flat within .005 (.13 mm) and have a .000032 (.0008 mm) finish or better.
p. Pitch dia. must accept a class 2 "GO" gage only.
r. Minor dia. must not be greater than .140 (3.56 mm).
s. Concentric with "BD" within .010 (.254 mm).
t. Axis of dia. "BD" to be located within .046 (1.17 mm) radius of true position.

FIGURE 1. Outline drawing of electron tube type DOD-029- Continued.

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